Universal Optimality of Dijkstra via Beyond-Worst-Case Heaps

论文评述

摘　要

Haeupler、Hladik、Rozhon、Tarjan 和 Tetek 的 *Universal Optimality of Dijkstra via Beyond-Worst-Case Heaps* 研究 Dijkstra 算法在 beyond-worst-case 视角下的最优性。本文评述以 2025 年 5 月 7 日发布的 arXiv v4 扩展版为准；该版本在 FOCS 2024 会议论文基础上重写并改进了若干证明。文章讨论的核心任务是 distance order problem：给定源点 s，输出顶点按照从 s 出发的真实距离非递减排列的顺序。这个任务弱于完整 SSSP，因为它不要求输出所有距离数值；但它保留了 Dijkstra 算法最关键的扫描顺序，也适合用比较次数和图拓扑本身的顺序自由度来衡量。

文章证明, 在合适的 priority queue 支持下, 普通 Dijkstra 在运行时间上已经达到 universal optimality；进一步处理图中的 bottleneck 后，Dijkstra 的两个扩展版本还能在比较次数上达到相应下界。除了下界和算法, 作者还构造了支持 `decrease-key` 的 working-set heap，使时间上界不依赖额外的数据结构假设。文章关注的并非继续改进一般图上的 worst-case 上界，而是解释固定拓扑如何影响 Dijkstra 的图相关复杂度。

1. 论文解决的问题

2. 为什么这个问题重要

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6. 关键技术细节

7. 技术贡献与局限

8. 总结

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